

Magnetic Fields — Paper B

MEDIUM • 40 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer every question in the space provided. Write in sentences — a list of single words scores badly on this topic.
- Every field line you draw must carry an arrowhead. A line without one does not score.
- Where a question asks for a number, that number must carry a unit.
- The mark for each question is shown in square brackets on the right.
- Reminder: outside a magnet the field lines run from north to south, like poles repel and unlike poles attract.

SECTION 1 – TESTING MATERIALS

- 1.** Meera tests six objects with a bar magnet: an iron nail, a length of copper wire, a steel spoon, an aluminium can, a plastic comb and a disc of pure nickel.

(a) List the objects that are attracted to the magnet. [2]

(b) Meera says her results prove that all metals are magnetic. Explain why her results show the opposite, naming two of her objects. [2]

- 2.** A brass key and a steel key look almost identical. Explain why only one of them is attracted to a magnet. [3]

SECTION 2 – DRAWING AND PLOTTING THE FIELD

- 3.** A bar magnet lies on the page with its north pole on the left.

- (a)** Draw the magnetic field pattern around it, using at least four lines. [3]

- (b)** State two things, other than the shape of the curves, that a marker looks for in your diagram. [2]

- 4.** Describe, step by step, how to plot a magnetic field line using a plotting compass. [3]
Your answer must say what you do with the compass and how the line ends up with an arrow on it.

- 5.** Arjun draws a field diagram in which two lines cross just above the north pole. [3]
Explain why this cannot be correct, and state what he should have drawn instead.

SECTION 3 – MAGNETS AT WORK

- 6.** The north pole of a bar magnet is touched to a steel paper clip. The clip sticks, and a second clip then hangs from the first.

(a) Explain why the first clip sticks to the magnet. [2]

(b) Explain why the second clip hangs from the first, even though the first clip is not a permanent magnet. [2]

7. The Earth behaves as though a huge bar magnet were buried along its axis.

(a) State what kind of magnetic pole lies at the geographic North Pole, and justify your answer. [2]

(b) Explain why a bar magnet hung freely on a thread always settles pointing roughly north to south. [2]

8. An object is attracted to a magnet.

(a) Explain why this does not prove the object is a magnet. [2]

(b) Describe the test that would prove it, and state what you would observe. [2]

End of paper. Before you hand it in, go back over every diagram and check that each field line carries an arrowhead pointing from north to south.